

Task Description | End-Sem Practical Examination | EE-A | Batch 2

Write a C program that accepts the **last 5 digits** of your own 11-digit roll number as an array of digits. The program should perform sorting, array reversal, and Stack operations based on the roll number. For example, if the roll number is **25295918011**, then, the last five digits are **18011** and it should be stored as {1, 8, 0, 1, 1}.

Tasks to be performed

1. Input the last 5 digits of the roll number into an integer array.
2. Choose and apply sorting algorithm (non-decreasing order)
 - a. Preserve the original copy of the array for later.
 - b. Convert the digit array into a complete 5-digit roll number and compute:
roll_number % 3
 - i. if roll_number % 3 == 0, use Selection Sort
 - ii. if roll_number % 3 == 1, use Insertion Sort
 - iii. if roll_number % 3 == 2, use Bubble Sort
3. Print the sorted 5-digit roll number. For example, **18011** will be printed as **01118** (without spaces)
4. Push the elements of the original copy of the roll number onto a Stack in the original order. For example, {1, 8, 0, 1, 1} will be pushed one by one in this order - **push(1), push(8), push(0), push(1), push(1)**.
5. Pop the stack one by one and print the elements in the reversed order. For the given example, this output will be **11081**.
6. Enqueue the popped elements one by one from the Stack to a Queue. For the given example, if the popping order is **1, 1, 0, 8, 1**, then the enqueue operations will also be in the same order - **enqueue(1), enqueue(1), enqueue(0), enqueue(8), enqueue(1)**.
7. Dequeue each element in the queue and print the final dequeued elements. For the given example, the output will be **11081**.

Important: Each print output must be on a new line. If the given example outputs do not contain whitespaces between digits, then your output must also not contain whitespaces.